← Back **Reinforcement learning introduction**
Graded Assignment • 30 min

Due Apr 3, 11:59 PM CEST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1 / 1 point

1. You are using reinforcement learning to control a four legged robot. The position of the robot would be its _____.

- return
- state
- reward
- action

Correct
Great!

1 / 1 point

2. You are controlling a Mars rover. You will be very very happy if it gets to state 1 (significant scientific discovery), slightly happy if it gets to state 2 (small scientific discovery), and unhappy if it gets to state 3 (rover is permanently damaged). To reflect this, choose a reward function so that:

- R(1) < R(2) < R(3), where R(1) and R(2) are negative and R(3) is positive.
- R(1) > R(2) > R(3), where R(1), R(2) and R(3) are positive.
- R(1) > R(2) > R(3), where R(1) and R(2) are positive and R(3) is negative.
- R(1) > R(2) > R(3), where R(1), R(2) and R(3) are negative.

Correct
Good job!

1 / 1 point

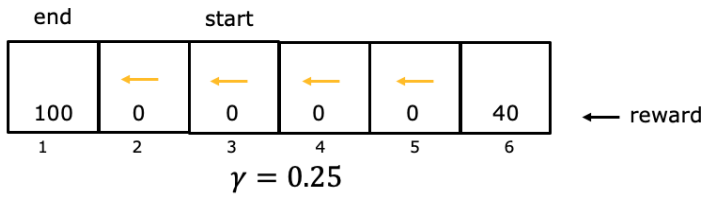
3. You are using reinforcement learning to fly a helicopter. Using a discount factor of 0.75, your helicopter starts in some state and receives rewards -100 on the first step, -100 on the second step, and 1000 on the third and final step (where it has reached a terminal state). What is the return?

- -0.25*100 - 0.25^2*100 + 0.25^3*1000
- -0.75*100 - 0.75^2*100 + 0.75^3*1000
- -100 - 0.25*100 + 0.25^2*1000
- -100 - 0.75*100 + 0.75^2*1000

Correct
Awesome!

1 / 1 point

4. Given the rewards and actions below, compute the return from state 3 with a discount factor of $\gamma = 0.25$.



- 0.39
- 25
- 6.25
- 0

Correct
If starting from state 3, the rewards are in states 3, 2, and 1. The return is
$0 + (0.25) \times 0 + (0.25)^2 \times 100 = 6.25$.